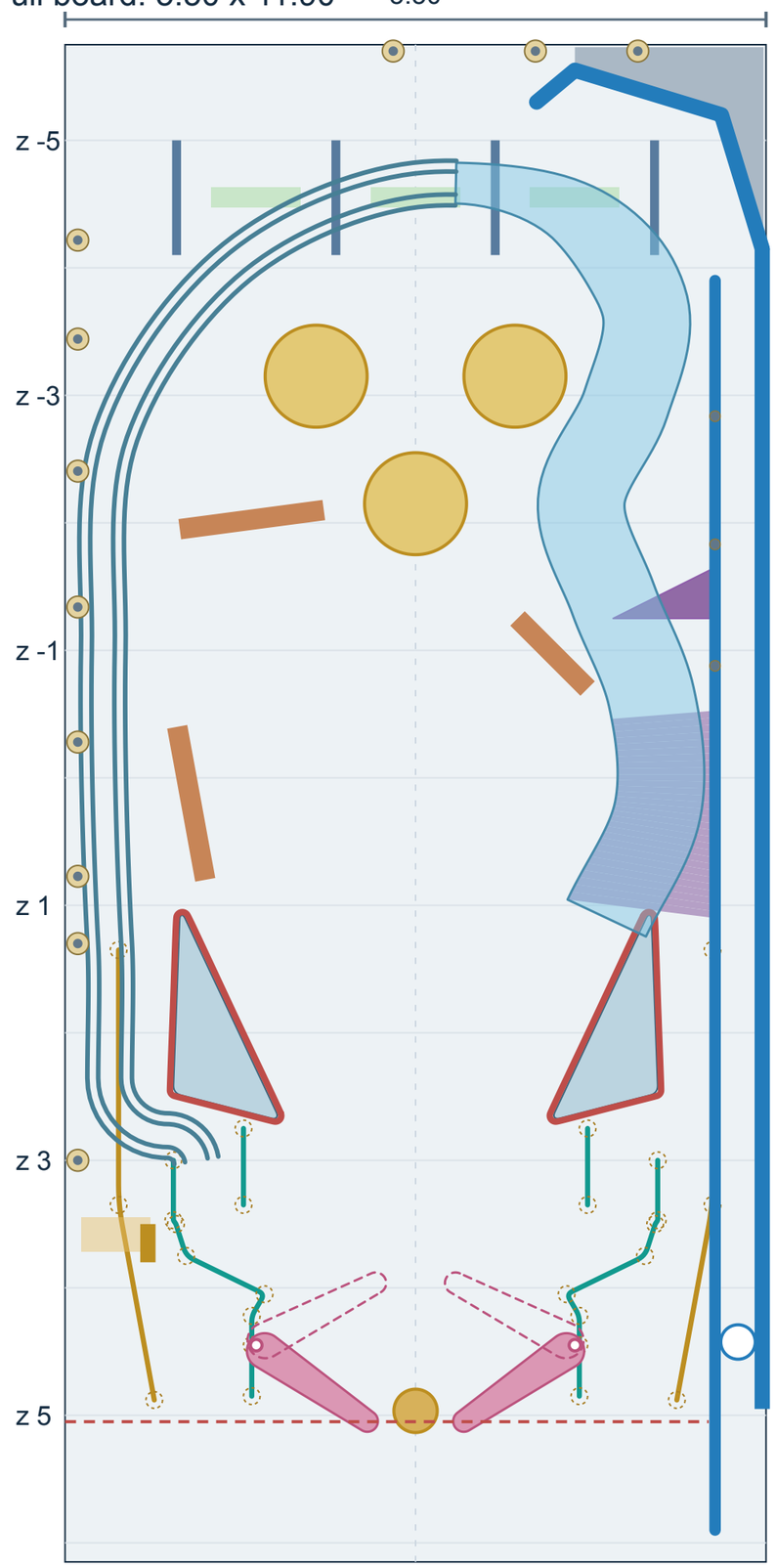
1 Accepted layout: scale, paths and relationships

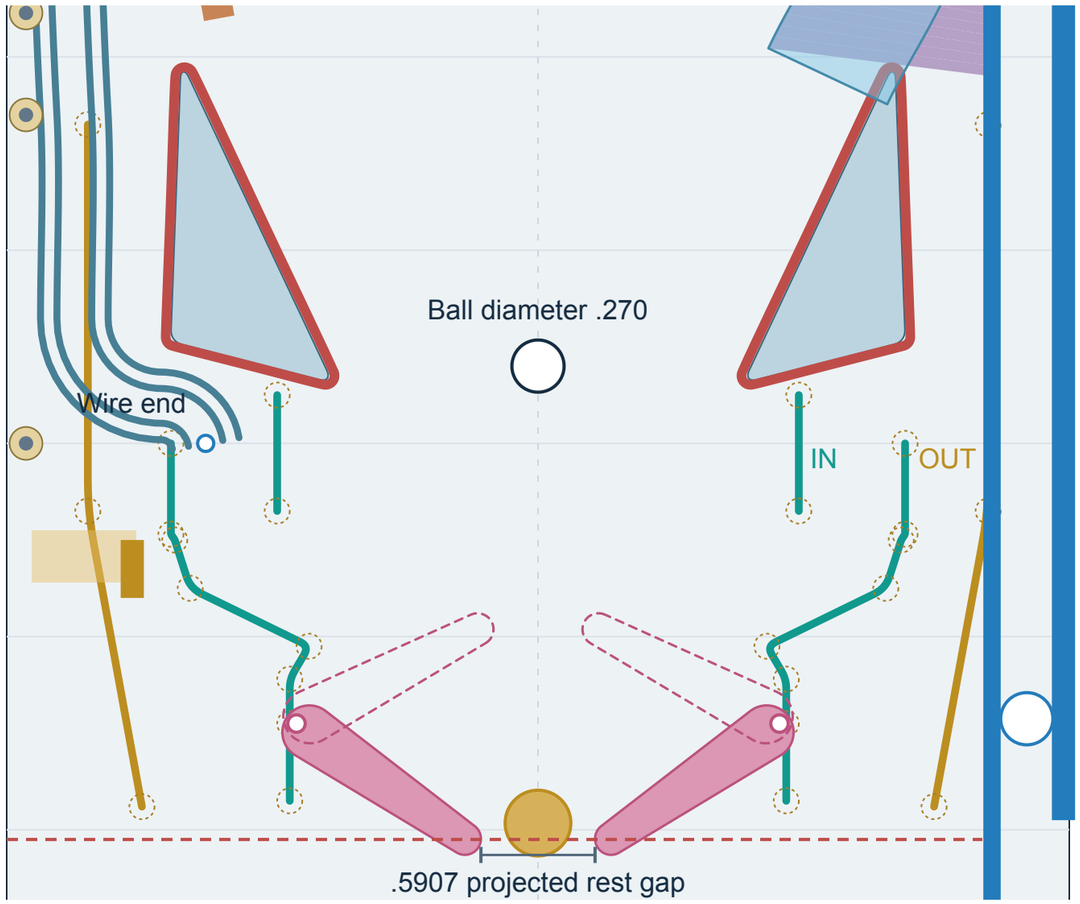
ACTUAL geometry projection. Elevated parts overlap ground parts in this view; this is not a free-passage certificate.

Full board: 5.50 x 11.90



Blue: shooter / pale cyan: elevated route
Purple: low base + nose / red: sling rubber
Ground direction +Z / front

Lower detail: actual broad sling, paired guides and bat sweep



Solid bats: rest. Dashed bats: held. Gold dotted circles: visible cap envelopes, not full passage clearance.

What the comparison actually establishes

MODEL: ball D .270; board is 20.37 balls wide and 44.07 balls long.

MODEL: sling rubber/contact reaches |x| 1.951968; broad cover is retained.

MODEL: straight triple-rail overlap z 3.00..3.35 = .350 = 1.30 ball diameters.

MODEL: printed IN/OUT lengths 1.10 / 1.70 = .647. Ink is not channel width.

MODEL: wire endpoint (-1.72, .44, 3.00); nominal supported center Y .575.

REFERENCE (perspective estimate only): broad slings, a right S and left wire, long near-parallel lower IN/OUT bands, readable hardware and a visible center gap.
Right band vertical-extent estimate IN/OUT ~1.1 (roughly .9..1.4).
The raised mouth may be an entry OR a return outlet. Image alone cannot decide.

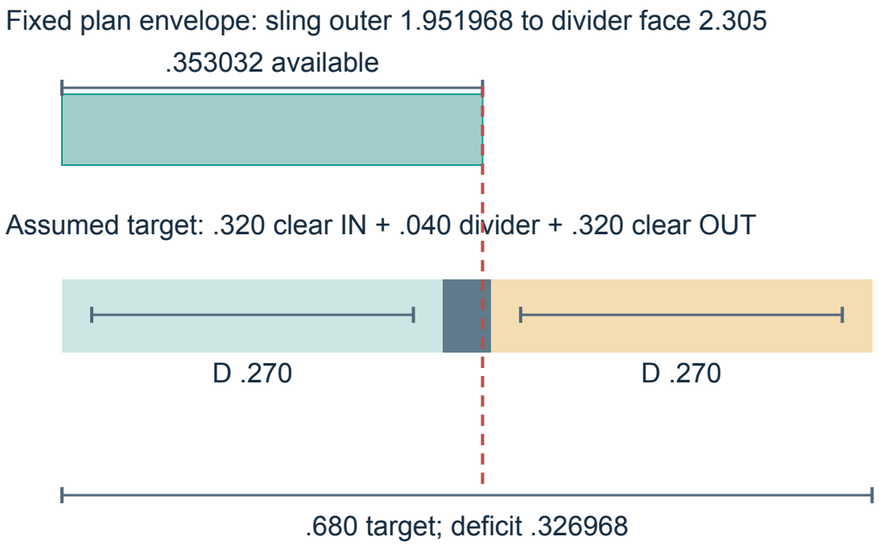
Reference ratio is not the same measurement as model ink length, and is not CAD.

SOURCE: accepted c1f7aeb at docs HEAD 4b3b4f0. Model units, not manufacturer dimensions. No new geometry or dynamic test.

2 Sections: conditional constraints, not universal impossibility

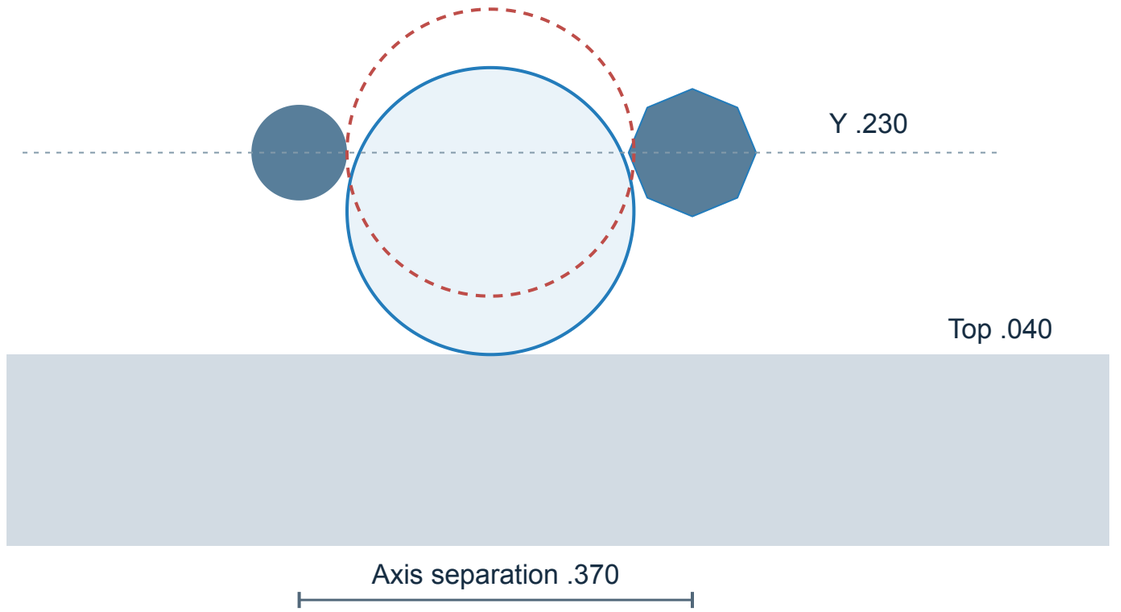
ACTUAL dimensions and explicit idealized calculations. Red means a failed condition; orange means a design assumption.

A Lower width allocation beside the broad sling



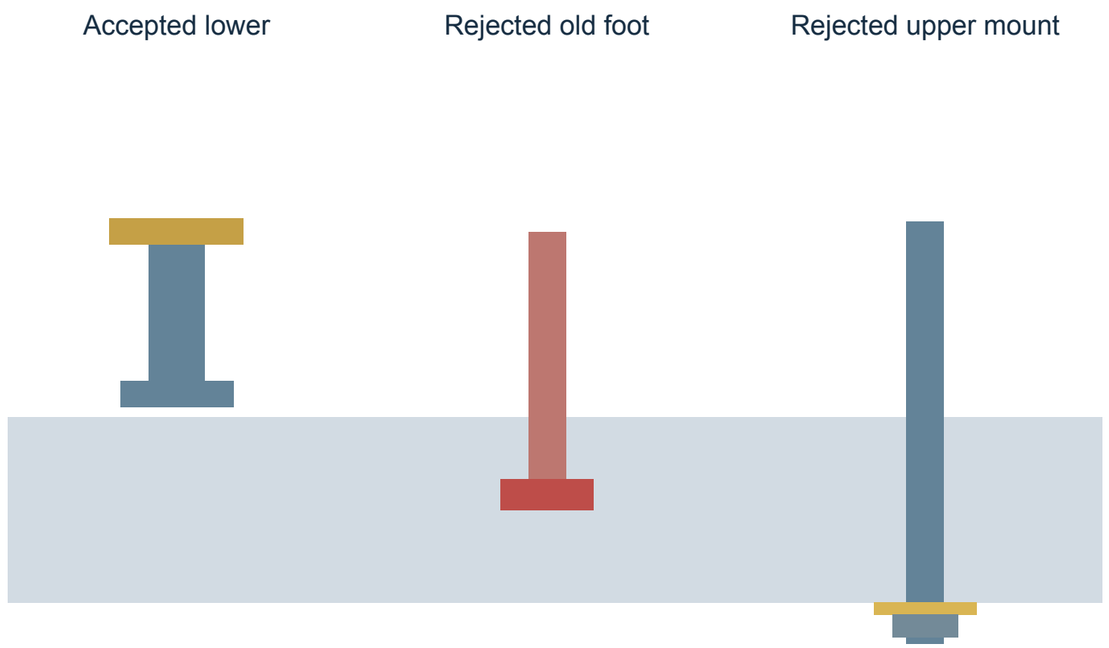
.320 = ball .270 + assumed .050 total spare, not a proven tolerance.
Even .270 + .040 + .270 needs .580 in this fixed PLAN allocation.
This rules out that allocation, not every 3D route or overall redesign.

B Shooter X-Y section: ground fit is not all-height fit



Blue: floor-center .175. Red dashed: axis-height .230.
Ideal circles: axis-height gap .265 < D .270; floor-center interval ~.01152.
Accepted outer is OCTAGONAL. Round-stock change adds up to .004567.

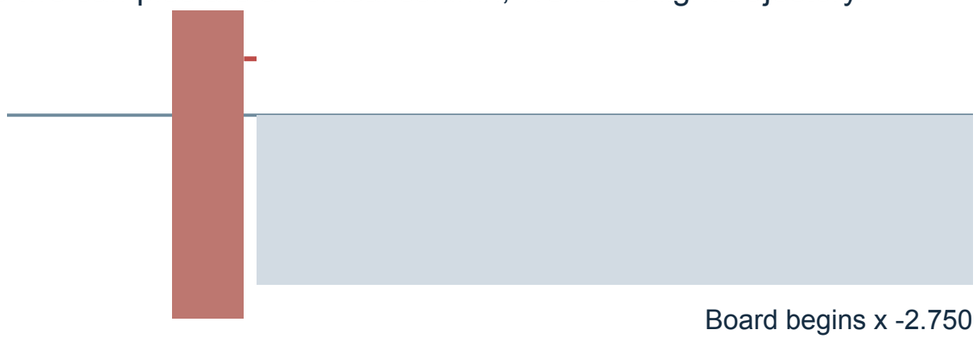
C Mount height: visible board Y [-.140, +.040]



Old foot [-.050, -.020] is inside the board, not under it.
Upper washer [-.152, -.140] and shaft to -.180 reach the underside.
Accepted lower caps are visible .130-wide parts; no dedicated cap collider.
No drawing here certifies holes, load paths or whole-candidate acceptance.

D Coupled handoffs and outside-board support

Wire end: (-1.72, .44, 3.00). Release direction and speed matter.
At the SAME XZ, rejected lower rail gaps were:
+.217768 at supported sphere-center Y .575; -.032706 at Y .175.
These two samples are a vertical column, NOT the flight trajectory.



Old relocated shaft x -2.800, r .037 ends at x -2.763:
a .013 lateral gap to the board. Longer shaft alone cannot attach.

OUT guide also shares the left earned-save region and firing axis.
Preserve real discharge; do not hide an exterior pocket with artwork.

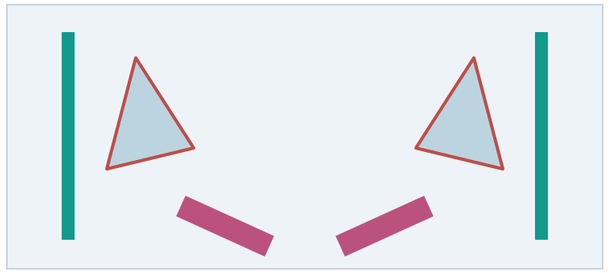
SOURCE: accepted c1f7aeb at docs HEAD 4b3b4f0. Model units, not manufacturer dimensions. No new geometry or dynamic test.

3 Whole-layout choices: concepts, not tested shapes

All three retain user-facing input/force behavior and must prove natural play plus equal-or-better continuous aim range.

A Keep the current play skeleton

Keep current shooter / roof / nose / bats.
Preserve terminal heel and wire handoff.
Explore earlier coupled IN + OUT transition.
Mounts and outside collection stay in scope.

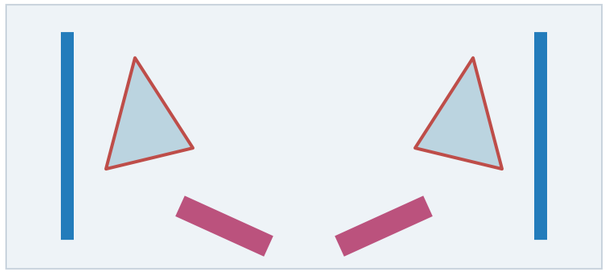


RELATIONSHIP SKETCH / NOT TO SCALE

Base: tested play; reference incomplete.
Any new transition is still unproved.
Limit: no promise of reference-long lanes.
Unknown: IN arrival / wire / save axis.
Reject: cosmetic-only elongation.

B Redistribute width as one system

Allocate width to both IN and OUT.
Co-design board, straight shooter and exit.
Include nose access, bars, sensors, supports.
Keep current bat / heel as a chosen baseline.

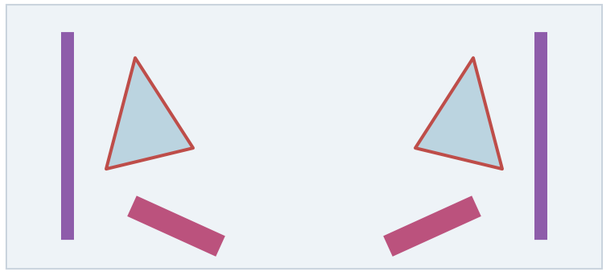


RELATIONSHIP SKETCH / NOT TO SCALE

Section room may exist; route is unproved.
Cost: entrance and lower return are coupled.
Unknown: all-height connector + natural feed.
Reject: rigid roof shift / repeat R1.70 wedge.

C Replan the mechanical topology

Allow pivot / heel / drain locations to change.
First compare current bat length (~1.14).
Do not silently require old center opening.
Decide if raised mouth is entry or return.



RELATIONSHIP SKETCH / NOT TO SCALE

Layout freedom, not proven viability.
Cost: save, difficulty, capture and aim requalify.
If mouth returns: another natural ascent needed.
No portal, hidden floor or automatic rescue.

Fixed outcomes vs selectable geometry

MUST KEEP: keyboard/touch semantics, force contract, natural weak/medium/strong feed, real IN capture then release/repress refire,
OUT drain and earned kickback, shared visible/contact boundaries, equal-or-better continuous aim coverage, reference quality and mobile.

NOT absolute user constraints: old pivot X/Z, exact center gap, board width/length, old shooter axis, or the reference mouth being an entry.
Illustration only: each pivot +.400 outward with the SAME bat/angle increases projected rest gap .590694 -> 1.390694.
The older ~1.625 bat calculation additionally FIXED the old opening; it is not a required bat length or an impossibility proof.

FIRST DECISION: choose route topology and which datums may move. No concept is qualified for adoption by this sheet.
Then: shared all-height corridor + reference views -> six natural powers / weak return -> IN capture/refire + OUT save ->
strong shots + original aim matrix -> full regression / controller games -> real phone and human play. Preserve every failed assertion.

No independent geometry improvement is currently qualified for adoption. A useful proposal is evidence/diagram tooling, not a cosmetic patch.

SOURCE: accepted c1f7aeb at docs HEAD 4b3b4f0. Model units, not manufacturer dimensions. No new geometry or dynamic test.